Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$M NH_4OH$ and $1 M HCl$ are mixed to make total volume $300\ mL$. If pH of the mixture $1\ s$ 9.26 and $pK_a\left(NH_4^+\right) = 9.26$ then volume ratio of NH_4OH and HCl will be:

(A) $1:1$

(B) $1:2$

(C) $2:1$

(D) $3:1$

Q.2

If the pK_a of a weak acid HA is 4.80 and the pK_b of a weak base BOH is 4.78. Then, the pH of an aqueous solution of the corresponding salt, BA will be

(A) 8.23

(B) 9.41

(C) 7.01

(D) 5.91

Q.3

The equlibrium constant of the reaction of a weak acid (HA) with a strong base $NaOH$) is 10^9 at $25^{\circ}C$. Hence K_b for A^- will be equal to

(A) 10^{-5}

(B) 10^{-9}

(C) 10^5

(D) 10^{-6}

Q.4

$50ml$ of $0.1M$ HA is titrated with $0.1M$ strong base (NaOH). The minimum value of equilibriwn constant (k) so that when $49.5ml$ of titrant has been added, the reaction between $HA \& OH$ is essentially complete and the pH changes by 2.00 units on addition of two more drops $(0.1\ mL)$ of titrant is

(A) 2×10^{10}

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(B) 2×10^9

(C) 5×10^8

(D) 2×10^7

Q.5

Calculate the ratio of pH of a solution containing $1\ \mathrm{mol}$ of $\mathrm{NaCN} + 1\ \mathrm{mol}$ of $\mathrm{HNO_3}$ per liter and of other solution containing lmol NaCN + lmol of HCN per liter
Given : $\mathbf{K_n}$ of HCN is the order of 10^{-10}.

(A) $1:1$

(B) $2:1$

(C) $1:2$

(D) Data insufficient

Q.6

pH of $0.005\mathrm{M}\mathrm{HCOOH}\left[\mathrm{K_a} = 2 \times 10^{-4}\right]$ is equal to

(A) 3

(B) 2

(C) 4

(D) 5

Q.7

Which of the following is strongest acid

(A) $\mathrm{pH} = 1$

(B) $\mathrm{pH} = 2$

(C) $\mathrm{pH} = 3$

(D) $\mathrm{pH} = 4$

Q.8

What volume of $0.80\mathrm{M}$ NaOH must be added to $50\ \mathrm{mL}\mathrm{M}$ solution of HA, a weak acid $\left(\mathrm{K_b}\right.$ for $A^- = 10^{-10}$ and $K_w = 10^{-14}\left.\right)$ so as to adjust the pH of the mixture at 4.6 ?

(A) $62\ \mathrm{mL}$

(B) $60\ \mathrm{mL}$

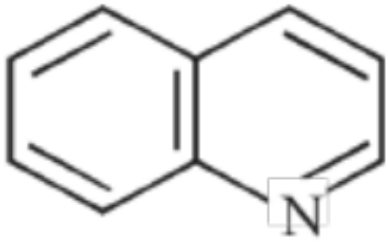
(C) $50\ \mathrm{mL}$

(D) $15\ \mathrm{mL}$

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Q.9

The congugate acid of the base quinoline has $pK_A = 4.88$. What frachon of molecules protonated in a $0.010M$ aqueous solution of quinoline ?



(A) 1.2×10^{-3}

(B) 2.8×10^{-4}

(C) 3.11×10^{-4}

(D) 1.87×10^{-4}

Q.10

If a $0.1M$ solution of HCN is 0.01% ionised, the ioniation constant for HCN is

(A) 10^{-3}

(B) 10^{-5}

(C) 10^{-7}

(D) 10^{-9}

Q.11

$\mathbf{pK}_{a1}$ and $\mathbf{pK}_{a_2}$ of H_2SO_3 are 1.82 and 7.2 respectively . The pH after $20\ mL$ of $0.16M$ $NaOH$ is added to $40\ mL$ of $0.08MH_2SO_3$ is :

(A) 7.0

(B) 1.82

(C) 8.8

(D) 4.51

Q.12

A solution of $0.1MNaZ$ has $pH = 8.90$. The K_a of HZ is

(A) 6.3×10^{-11}

(B) 6.3×10^{-10}

(C) 1.6×10^{-5}

(D) 1.6×10^{-6}

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Q.13

A solution contains $10\ \mathrm{mL} 0.1\ \mathrm{N} \mathrm{NaOH}$ and $10\ \mathrm{mL}$ $0.05\ \mathrm{N} \mathrm{H_2SO_4}$, pH of this solution is :

(A) less than 7

(B) 7

(C) zero

(D) greater than 7

Q.14

Assuming that the degree of hydrolysis is small, the pH of $0.1\mathrm{M}$ solution of sodium acetate $\left(\mathrm{K_a} = 1.0 \times 10^{-5}\right)$ will be

(A) 5.0

(B) 6.0

(C) 8.0

(D) 9.0

Q.15

Which of the following statements about pH and $\mathrm{H^+}$ ion concentration is incorrect?

(A) Addition of one drop of concentrated HCl in $\mathrm{NH_4OH}$ solution decreases pH of the solution.

(B) A solution of the mixture of one equivalent of each of $\mathrm{CH_3COOH}$ and NaOH has a pH of 7

(C) pH of pure neutral water is not zero

(D) A cold and concentrated $\mathrm{H_2SO_4}$ has lower $\mathrm{H^+}$ ion concentration than a dilute solution of $\mathrm{H_2SO_4}$

Q.16

If the reaction between $\mathrm{CO_2}$ and $\mathrm{H_2O}$ is $\mathrm{CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-}$
If $\mathrm{CO_2}$ escapes from the system

(A) pH will decrease

(B) $\mathrm{H^+}$ concentration will decrease

(C) $\mathrm{H_2CO_3}$ concentration will be altered

(D) The forward reaction is promoted

Q.17

Calculate the pH of a solution containing $0.1\mathrm{M}\ \mathrm{HCO_3^-}$ and $0.2\mathrm{M} \mathrm{CO_3^{2-}}$

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$\left[\mathrm{K}_1\left(\mathrm{H}_2\mathrm{CO}_3\right)=4.2 \times 10^{-7} \times 10\right.$ and
$\left.\mathrm{K}_2\left(\mathrm{HCO}_3^{-}\right)=4.8 \times 10^{-11}\right]$

(A) 3.18

(B) 10.62

(C) 6.62

(D) 9.31

Q.18

What is the pH of a $10^{-4} \mathrm{MOH}^{-}$ solution at $330\ \mathrm{K}$, if K_{w} at $330\ \mathrm{K}$ is $10^{-13.6}$?

(A) 4

(B) 9.0

(C) 10

(D) 9.6

Q.19

$20\ \mathrm{mL}$ of $0.2 \mathrm{MNaOH}$ are added to $50 \mathrm{mL}$ of
$0.2 \mathrm{MCH}_3\mathrm{COOH}\left(\mathrm{K}_{\mathrm{a}}=1.8 \times 10^{-5}\right)$ the pH of the solution is

(A) 4.56

(B) 4.73

(C) 9.45

(D) 6.78

Q.20

NaOH is a strong base. What will be pH of $5.0 \times 10^{-2} \mathrm{MNaOH}$ solution $?(\log 2=0.3)$

(A) 14.00

(B) 13.70

(C) 13.00

(D) 12.70

Answers & Solutions

Q.1 Answer:

$2 : 1$

Solution:

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$\mathrm{pH} = 9.26$ **indicates** $[\mathrm{NH_4OH}] > [\mathrm{HCl}]$ **and thus mixture is a buffer since** HCl **will react with equivalent amount of** $\mathrm{NH_4OH}$ **forming** $\mathrm{NH_4Cl}$ **Let** $\mathrm{HCl} = \mathrm{xmL} = \mathrm{x}$ **millimol**

$\mathrm{NH_4OH} = (300 - \mathrm{x})\mathrm{mL} = (300 - \mathrm{x})$ **millimol**

$\mathrm{NH_4Cl}$ **formed** $= \mathrm{x}$ **millimol**

$\mathrm{NH_4OH}$ **unreacted** $= 300 - \mathrm{x} - \mathrm{x} = (300 - 2\mathrm{x})$ **millimol**

$\mathrm{pOH} = 14 - 9.26 = 4.74 \quad \mathrm{pK_b} = 14 - 9.26 = 4.74$

$\mathrm{pOH} = \mathrm{pK_h} + \log_{[\mathrm{NH_4OH}]}$

$4.74 = 4.74 + \log \frac{x}{(300-2x)} \frac{x}{300-2x} = 1$

$\therefore \mathrm{x} = 100\ \mathrm{mL} =$ **volume of** $\mathrm{HCl}(300 - \mathrm{x})$

$= 200\ \mathrm{mL} =$ **volume of** $\mathrm{NH_4OH}$

Hence 2: 1

Q.2 Answer:

7.01

Solution:

$\mathrm{BOH} + \mathrm{HA} \rightarrow \mathrm{B^+} + \mathrm{A^-} + \mathrm{H_2O}$

$\mathrm{pH} = \frac{1}{2}\ \mathrm{pK_w} + \frac{1}{2}\ \ \mathrm{pK_a} - \frac{1}{2}\mathrm{pK_b}$

$= \ \ \frac{1}{2}14 + 4.80 - 4.78 = 7.01$

Q.3 Answer:

10^{-9}

Solution:

$\mathbf{HA} \rightleftharpoons \mathbf{H^+} + \mathbf{A^-}, \mathbf{K}$

$\mathbf{H^+} + \mathbf{OH^-} \rightleftharpoons \mathrm{H_2O}, \frac{1}{\mathbf{K_w}}$

$\mathrm{HA} + \mathrm{OH^-} \rightleftharpoons \mathrm{H_2O} + \mathbf{A^-}, \mathbf{K_b} = \frac{\mathbf{K_a}}{\mathbf{K_w}} = 10^9$

$\frac{\mathrm{K_a}}{10^{-14}} = 10^9$ **so** $\mathrm{K_a} = 10^{-5}$

$\therefore \mathbf{K_g K_b} = \mathbf{K_w}, , \mathbf{K_b} = \frac{\mathbf{K_N}}{\mathbf{K_n}} = \mathbf{10^{-9}}$

Q.4 Answer:

2×10^9

Solution:

The $\mathrm{pH}0.05\ \mathrm{mL}$ **beyond the equivalence point can be calculated as follows**

$\left[\mathrm{OH^-}\right] = \frac{0.05 \times 0.1}{100.05} = 5 \times 10^{-5}\mathrm{M}$

$\mathrm{POH} = 4.3 \quad \mathbf{pH} = 9.7$ **before**

If $\Delta\mathrm{pH} = 2$ **units, the** $\mathrm{pH}0.05\ \mathrm{mL}$ **before the equivalence point must be** 7.7**. At this point if the reaction is complete, we have only** 0.005 **m,mol of HA unreacted,**

Hence

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$\mathbf{pH = pk_a + \log \frac{[A^-]}{[HA]}} \quad 7.7 = \mathrm{pk_a} + \log \frac{4.995}{0.005}$

$pk_a = 4.7 k_a = 10^{-4.7} = 2 \times 10^{-5} \quad k = \frac{k_a}{k_w} = \frac{2 \times 10^{-5}}{10^{-14}} = 2 \times 10^9$

Q.5 Answer:

$1 : 2$

Solution:

K_a of HCN is the order of 10^{-10} so at this concentration, a of HCN will be negligible.

$$\mathrm{NaCN + HNO_3 \rightarrow HCN + NaNO_3}$$

$$1 \qquad\qquad 1$$

$$0 \qquad\qquad 0 \qquad\qquad 1 \qquad\qquad 1$$

$[\mathrm{HCN}] = 1\mathrm{M}$

$\mathrm{pH_1} = \frac{1}{2}\mathrm{pK_a} - \frac{1}{2}\log 1$

$\mathbf{pH_2 = pK_a}$

$\frac{\mathrm{pH_1}}{\mathrm{pH_2}} = \frac{1}{2}$

Q.6 Answer:

3

Solution:

Since C is low and K is high, we will use the following formula:

$\mathrm{K_a} = \frac{\mathrm{C}\alpha^2}{1-\alpha}$

$\Rightarrow 2 \times 10^{-4} = \frac{(0.005)\alpha^2}{1-\alpha}$

$\Rightarrow \alpha = 0.18$

$\Rightarrow \mathrm{pH} = -\log \mathrm{C}\alpha$

$\Rightarrow -\log(0.05 \times 0.18) \approx 3$

Q.7 Answer:

$\mathrm{pH} = 1$

Solution:

$\mathbf{pH} = -\log[\mathrm{H^+}]$

$\Rightarrow 1 = -\log\left[\mathrm{H^\dagger}\right] \Rightarrow [\mathrm{H^+}] = 10^{-1}$ **which is higher than $10^{-2}, 10^{-3}$ and 10^{-4}**

$\therefore$ Options , and are wrong.

Q.8 Answer:

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$50\ \mathrm{mL}$

Solution:

Let, v mL of $0.80\mathrm{M}$ NaOH be added then

No. of m mole of salt (NaA) formed $= 0.8v$

No. of m mole of acid remaining unreacted $= 50 - 0.8\mathrm{v}$

$\mathrm{K_a}$ **for** $\mathrm{HA} = 10^{-14}/10^{-10} = 10^{-4}$

$\therefore\ \mathrm{pK_a} = 4$

$4.6 = 4 + \log \frac{0.8v}{50-0.8v}$

$\therefore\ \frac{0.8v}{50-0.8v} = 4,$ **So** $v = 50\ \mathrm{mL}$

Q.9 Answer:

2.8×10^{-4}

Solution:

$\mathbf{pK_a} = 4.88 \Rightarrow \mathrm{pK_b} = 14 - 4.88 = 9.12$

or $\mathrm{K_b} = 10^{-9.12} = 7.6 \times 10^{-10}$

Aqueous solution of quinoline establish following equilibrium.

$$\mathbf{Q + H_2O \rightleftharpoons Q^+ \rightleftharpoons Q + OH}$$

(Quinoline)

$\quad \mathrm{Initial} \quad \mathbf{0.0}10$

at equilibrium $0.010 - x\ x \quad x$

$K_b = \frac{x^2}{0.010-x} x = -2.8 \times 10^{-6}$

$\mathrm{f} = \frac{\left[\mathrm{QH^+}\right]}{[\mathrm{Q}]_{\mathrm{intial}}} = \frac{2.8\times10^{-6}}{0.010} = 2.8 \times 10^{-4}$

Q.10 Answer:

10^{-9}

Solution:

$C = 0.1\ \mathrm{m} \quad \alpha\% = 0.01$

$\alpha = \frac{0.01}{100} = 1 \times 10^{-4}$

$\mathrm{K} = \mathrm{C}\alpha^2 = 0.1 \times \left(1 \times 10^{-4}\right)^2$

$= 1 \times 10^{-9}$

Q.11 Answer:

4.51

Solution:

$\underset{40\times0.08\mathrm{meq}}{\mathrm{H_2SO_3}} + \underset{20\times0.16\mathrm{meq}}{\mathrm{NaOH}} \longrightarrow \mathrm{NaHSO_3} + \mathrm{H_2O}$

$\therefore$ **at first equivalence** $\mathrm{pH} = \frac{\mathrm{PK_{a_1}} + \mathrm{PK_{a_2}}}{2}$

$= \frac{1.82+7.2}{2} = 4.51$

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Q.12 Answer:

1.6×10^{-5}

Solution:

NaZ **is salt of** $\mathrm{W_A/S_B}$

$\therefore \mathrm{pH} = \frac{1}{2}\left(\mathrm{pK_w} + \mathrm{pK_a} + \log \mathrm{C}\right)$

$8.9 \times 2 = 14 + \mathrm{pK_a} + \log 0.1$

$17.8 = 14 + \mathrm{pK_a} - 1$

$\mathrm{pK_a} = 4.8$

$\mathrm{K_a} = \mathrm{Antilog}(-4.8)$

$= 1.585 \times 10^{-5} \approx 1.6 \times 10^{-5}$

Q.13 Answer:

7

Solution:

Milliequivalents of $\mathrm{NaOH} = 10 \times 0.1 = 1$ **Milliequivalents of** $\mathrm{H_2SO_4} = 10 \times 0.05 = 0.5$

$2\mathrm{NaOH} + \mathrm{H_2SO_4} \longrightarrow \mathrm{Na_2SO_4} + \mathrm{H_2O}$

2 equ. 1 equ.

$\therefore$ **1 equivalent of** NaOH **reacts with** $0.5\mathrm{eq}$

of $\mathrm{H_2SO_4}$ **to give neutral** $(\mathrm{pH} = 7)$ **solution.**

Q.14 Answer:

9.0

Solution:

Sodium acetate is a salt of strong base and weak acid.

$\therefore \mathrm{pH} = 7 + \frac{1}{2}\mathrm{p}K_a + \frac{1}{2}\log c$

where $\mathrm{p}k_a = -\log K_a = -\log 10^{-5} = 5$

$\log c = \log 10^{-1} = -1$

$\mathrm{pH} = 7 + \frac{5}{2} - \frac{1}{2} = 9.0$

Q.15 Answer:

A solution of the mixture of one equivalent of each of $\mathrm{CH_3COOH}$ and NaOH has a pH of 7

Solution:

$\mathrm{CH_3COOH}$ **is weak acid while** NaOH **is strong base, so one equivalent of** NaOH **can not be neutralized with one equivalent of** $\mathrm{CH_3COOH}$**. Hence the solution of one equivalent of each does not have** pH **value as 7 . Its** pH **will be towards basic side as** NaOH **is a strong base hence conc. of** OH **will be more than the conc. of** $\mathrm{H^+}$

Q.16 Answer:

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H^+ concentration will decrease

Solution:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$
$$\rightleftharpoons H^+ + HCO_3^-$$

If CO_2 escapes, the equilibrium will shift to LHS and $[H^+]$ concentration will decrease

Q.17 Answer:

10.62

Solution:

$$HCO_3^- \rightarrow H^+ + CO_3^{2-}$$
$$K_2 = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]} = 4.8 \times 10^{-11}$$
$$[H^+] = \frac{4.8 \times 10^{-11}[HCO_3^-]}{[CO_3^{2-}]}$$
$$= 4.8 \times 10^{-11}(0.1/0.2)$$
$$pH = -\log[H^+]$$
$$= -\log\left(4.8 \times 10^{-11} \times 0.5\right) = 10.62$$

Q.18 Answer:

9.6

Solution:

$$K_w = 10^{-13.6}$$

i.e. $p\,K_w = pH + pOH$

$\because pOH = -\log[OH^-]$

Given at 330 K

$$13.6 = pH + pOH$$
$$pOH = -\log 10^{-4}$$
$$pOH = 4$$
$$\therefore pH = 13.6 - 4 = 9.6$$

Q.19 Answer:

4.56

Solution:

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$NaOH$	+	CH_3COOH	$\rightleftharpoons$	CH_3COONa	+	H_2O
20×0.2		50×0.2		0		0
4		10		0		0
0		6		4		4

Conc. $= \frac{\text{Millimoles}}{\text{Total Volume}}$

$[CH_3COOH] = \frac{6}{70}; [CH_3COONa] = \frac{4}{70}$

$\text{pH} = -\log 1.8 \times 10^{-5} + \log \frac{4/70}{6/70}, \text{pH} = 4.56$

Q.20 Answer:

12.70

Solution:

Given $\left[OH^-\right] = 5 \times 10^{-2}$

$\therefore \text{pOH} = -\log 5 \times 10^{-2}$

$= -\log 5 + 2\log 10 = 1.30$

$\because \text{pH}+\text{pOH} = 14$

$\because \text{pH} = 14 - \text{pOH}$

$= 14 - 1.30 = 12.70$

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